

NUMERICAL ANALYSIS

PROBLEM SET 5

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1. PROBLEM 1

Derive the trapezoid integration scheme as an Adams-Moulton method with $m = 1$ step:

$$\frac{w_{j+1} - w_j}{h} = \frac{1}{2}[f(t_{j+1}, w_{j+1}) + f(t_j, w_j)]$$

to approximate the initial value problem $y' = f(t, y)$ for $a \leq t \leq b$ with initial condition $y(a) = \alpha$. Also derive the associated error term. Let $w_j \approx y_j := y(t_j)$.

- (a) Use the Lagrange form of the polynomial interpolant to write

$$f(t, y(t)) = P_1(t) + R_1(t).$$

Integrate $P_1(t) = L_{1,0}(t)f(t_{j+1}, y_{j+1}) + L_{1,1}(t)f(t_j, y_j)$ from $t = t_j$ to $t = t_j + 1$.

- (b) Set $w_{j+1} = w_j + \int_{t_j}^{t_{j+1}} f(t, y(t))dt$, approximating the 2nd term with $\int_{t_j}^{t_{j+1}} P_1(t)dt$, and derive the method above.
- (c) The error term $R_1(t) = \frac{f''(\xi, y(\xi))}{2}(t - t_{j+1})(t - t_j)$ for f (some $\xi \in (t_j, t_{j+1})$) can be integrated from $t = t_j$ to $t = t_{j+1}$ to get the error associated with the Adams-Moulton scheme:

$$\text{Error} = Mh^3 f''(\mu, y(\mu)).$$

Do this integration to find value of M .

- (d) What is the order of this one-step implicit method?

2. PROBLEM 2

Consider the initial value problem

$$y' = \frac{4t\sqrt{1+y^2}}{y}, y(0) = 1$$

for $0 \leq t \leq 5$.

Challenge. Show that $y(t) = \sqrt{(2t^2 + \sqrt{2})^2 - 1}$ is a solution to the initial value problem. Is it unique?

Numerically implement the following methods below, and run with enough values of step-size h to demonstrate that each of has convergence of global discretization error with rate $\mathcal{O}(h^2)$.

- (a) Optimal RK2 method
- (b) Two-step Adams-Bashforth
- (c) One-step Adams-Moulton method
- (d) Heun method (predictor-corrector)

Plot the error versus the step size on a log-log plot for all methods on the same axes to compare. Comment on the advantages and disadvantages of each.

Solution.

□

3. PROBLEM 3

Consider the linear multi-step method to approximate the initial value problem $y' = f(t, y)$ for $a \leq t \leq b$ with the initial condition $y(a) = \alpha$:

$$w_{j+1} - \frac{18}{11}w_j + \frac{9}{11}w_{j-1} + \frac{2}{11}w_{j-2} = \frac{6}{11}hf(t_{j+1}, w_{j+1}).$$

- (a) Is the method stable? If so, is it strongly or weakly stable? Show your work by considering the roots of the characteristic polynomial, noting that $\lambda = 1$ is a root, and you can even factor and write $P(\lambda) = (\lambda - 1) * (\dots)$.
- (b) Is the method convergent?
- (c) Is the method consistent?

Solution.

□

4. PROBLEM 4

Use a built-in ODE solver to numerically integrate the Lorenz equations

$$\begin{aligned}\dot{x} &= \sigma(x - y) \\ \dot{y} &= rx - y - xz \\ \dot{z} &= xy - bz\end{aligned}$$

with the following values for coefficients $\sigma = 10, r = 28, b = 8/3$.